

Building Community-Engaged Statistics and Data Science (CESDS) Courses

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GETTING STARTED

Warning

This handbook is in development.

This handbook provides resources for developing and teaching community-engaged statistics and data science (CESDS) courses at the undergraduate level.

We define **community-engaged statistics and data science (CESDS)** courses as...

The handbook covers both the development and implementation of CESDS courses, including course design, project scoping, community partnerships, student project work, ethical considerations, common challenges, and project handoff and sustainability.

The handbook is being developed during the 2026-2027 academic year by a working group. More information about the project and working group is available on the [CESDS website](#). Check back to learn more!

Acknowledgments

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Tips for Authors

This section is for contributors and will be removed from the final handbook.

This handbook is built using [Quarto](#). Below are some Quarto features that may be useful as you add content.

Headings

Use headings to organize sections within your chapter. They will appear automatically in the page table of contents.

```
## Main section
```

```
### Subsection
```

Citations

Use `@citationkey` to cite sources and add the corresponding BibTeX entry to `references.bib`.

```
This approach has been used in previous work @knuth84.
```

Links

Create descriptive links rather than displaying a full URL.

Learn more from the [RIOS Institute] (<https://qubeshub.org/community/groups/ri>)

Callouts

Callouts can be used for examples, notes, or other information you want to distinguish from the main text.

See examples of call-outs below.

```
::: {.callout-note}  
Add information here.
```

Cross-references

Add an ID to a section.

```
## Learning Objectives {#sec-learning-objectives}
```

Then link to it elsewhere using, see @sec-learning-objectives.

Comments

Use HTML comments for notes between authors that should not appear in the rendered handbook:

```
<!-- TODO: Add an example here. -->
```

Note

Add information here.

Warning

Add cautionary tale here.

Learn more about types of Callout Blocks at quarto.org. They can also be collapsible.

Example

Add an example here.

Supplemental or sidenote information can be placed in the margin.

Add a perspective here.

Part I

DEVELOPMENT

1 Course Design

1.1 Examples/types of CESDS courses?

1.2 Learning objectives

1.3 Sample Syllabi

1.4 Timelines

1.5 Assignments & Assessment

- Types of assignments (syllabus-like)
- Reference to specific assignments/assessment mentioned elsewhere in the Handbook
- Identifying resources (books, software, etc)

2 Scoping Projects

2.1 What makes a good project?

2.2 Data availability

2.3 Deliverables

(slides, presentation, paper, etc.)

3 Community Partnerships

3.1 Finding partners

3.2 Communication (sample emails)

3.3 Roles and expectations + scoping the project

3.4 Agreements, data use, etc.

Part II

IMPLEMENTATION

4 Student Project Work

4.1 Onboarding/engaging students

(ASCCR)

4.2 Teams/roles

(contracts)

4.3 Communication & Meetings

(consulting skills?)

4.4 Communicating to a variety of audiences

4.5 Project management

(Trello, GitHub, etc.)

4.6 Scaffolding

(project pieces, due dates, etc.) – Statistical Analysis Plan (SAP)

5 Ethics & Community Engagement

5.1 Power & positionality

5.2 Training

(background knowledge, conduct, privacy, CITI trainings etc)

5.3 Ethical data practice

5.4 Responsible communication

5.5 AI Considerations

6 Common Challenges

6.1 Partner stops responding

6.2 Data aren't available / aren't usable

6.3 Partner has a change in staff

6.4 Student group dynamics and workload issues

6.5 Challenging student approaches

(lack of professionalism, disengagement)

7 Wrap-up, Handover, & Sustainability

7.1 Deliverables, Documentation, & Reproducibility

7.2 Running a public event?

(Data walks, community meeting, etc)

7.3 Sustainability

7.4 Handoff checklist + Partner Feedback + Maintaining relationship

References